(iii) 1000,400,160 3. 1 4. -18 5.(i) 12 (ii) 7 6. $5\times(3^{11})$

7. 3072 9. $\frac{9}{2}$, 3, 2 (or) 2, 3, $\frac{9}{2}$ 10. ₹ 76577 11. ₹23820,₹24040

Exercise 2.8

1.(i) $\frac{25}{8}\left[1-\left(-\frac{3}{5}\right)^n\right]$ (ii) $\frac{1024}{3}\left[1-\left(\frac{1}{4}\right)^n\right]$ 2. 1820 3. 12

4.(i) $\frac{27}{2}$ (ii) 63 5. $\frac{1}{4}$ 6.(i) $\frac{4}{9}n-\frac{4\left[1-\left(\frac{1}{10}\right)^n\right]}{81}$

(ii) $\frac{10(10^n-1)}{27}-\frac{n}{3}$ 7. 3069 8. ₹ 174760 9. $\frac{41}{333}$

Exercise 2.9

1.(i) 1830 (ii) 1584 (iii) 3003 (iv) 1240 (v) 3256

(vi) 42075 (vii) 1296 2. 105625 3. 210

4. 15 5. 9 6. 4615 cm^2 7.(i) $4n^3+3n^2$ (ii) 2240

Exercise 2.10

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
(C)	(A)	(B)	(C)	(D)	(A)	(D)	(C)	(A)	(C)	(C)	(D)	(B)	(B)	(C)

Unit exercise-2

2.(i) 35 litres (ii) 5 (iii) 3 3. 1 6. -78

8. ₹1200 9. $\sqrt{2}$, $\sqrt{6}$, $3\sqrt{2}$, … 10. ₹27636

Exercise 3.1

1.(i) 2, -1, 4 (ii) $\frac{1}{2}, \frac{1}{3}, \frac{1}{4}$ (iii) 35, 30, 25

2.(i) infinitely many solution (ii) no solution (iii) unique solution

3. 24 years, 51 years, 84 years 4. 137 5. 7, 3, 2

Exercise 3.2

1.(i) x^2+2x-3 (ii) x^2+1 (iii) $x(x^2+4x+4)$ (iv) $3(x^2+1)$

2.(i) $8x^3y^2$ (ii) $-36a^3b^2c$ (iii) $-48m^2n^2$ (iv) $(p-1)(p-2)(p+2)$

(v) $4(x+3)(2x+1)(x-3)$ (vi) $2^3x^2(2x-3y)^3(4x^2+6xy+9y^2)$

Exercise 3.3

1.(i) $7xy$, $105x^2y^2$ (ii) $(x+1)$, $(x-1)(x+1)(x^2+x+1)(x^2-x+1)$

(iii) $x(x+y)$, $xy(x+y)$ 2.(i) $(a+6)(a-2)(a-3)$ (ii) $x(x-3a)^2(x^2+3ax+9a^2)$

3.(i) $4x^2(x-1)$ (ii) x^2-xy+y^2 4. (i) $(a+2)(a-7)$ (ii) $(x^2-y^2)(x^2+xy+y^2)$

Exercise 3.4

1.(i) $\frac{x-1}{x}$ (ii) $\frac{x-9}{x-2}$ (iii) $\frac{9}{x-1}$ (iv) $\frac{p+5}{2p(p-4)}$

2.(i) -5, 5 (ii) 2, 3 (iii) 1 (iv) 0, -3, 2

10th_Maths_Answer_English.indd 337 12/14/2024 4:13:52 PM